

ECON-7102: Environmental Economics I

Class 7: Instrument Choice (ii)

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Today's agenda

1. Questions?
2. Instrument choice
 - Environmental policy with competitive output markets
3. For next class...

Environmental policy with competitive output markets

- Until now, we've considered very simplified models of firm responses to environmental policy
 - Namely, emissions and output are separable.
 - Important to consider: how policy affects firms' cost structure, more realistic versions of command-and-control policies, how output prices factor into firms' production/abatement decisions, economic efficiency in the short vs. long run (i.e., with entry/exit into markets).
- All of this is to explore whether efficiency properties of different policy instruments from our simplified model carry over to richer models of firm behavior.
- *Competitive markets:*
 - All firms are profit maximizers
 - Firms have no market power (they are price takers)
 - Zero profit in the long run.

Competitive output markets

Preliminaries

- Firm's production technology for goods:

$$x = f(l_1, \dots, l_K)$$

- x is the level of output produced
- $l_1, \dots, l_K$ is a vector of inputs

- Firm's emission technology:

$$e = g(l_1, \dots, l_K)$$

- e is the level of emissions.

- Shape of f and g is not constrained (i.e., marginal products could be $+$, 0 , or $-$).

Competitive output markets

Firm's cost function (Case 1: when abatement is possible)

Cost function given by (where w_k are factor input prices):

$$C(x, e) = \min_{l_1, \dots, l_K} \left\{ \sum_{k=1}^K w_k l_k + \lambda[x - f(l_1, \dots, l_K)] + \mu[e - g(l_1, \dots, l_K)] \right\}$$

Assumptions:

- $C(x, e)$ is twice continuously differentiable, with $C_x(\cdot) > 0$ (marginal costs are positive)
- For any x there is an emissions level $\hat{e}^x$ such that $C_e(x, \hat{e}^x) = 0$ ($\exists$ a cost-minimizing emissions level w/o env. reg.)
- $C_e(x, e^x) < 0 \forall e < \hat{e}^x$ (moving below unregulated optimum increases firm's costs);
- $C_e(x, e^x) \geq 0 \forall e \geq \hat{e}^x$ (moving above unregulated optimum weakly increases firm's costs);
- $C_{xe} = C_{ex} < 0 \forall e < \hat{e}^x$
 - (MC of production is smaller if emissions level is allowed to rise)
 - (or, MAC curve shifts up if output level rises)

Competitive output markets

Firm's cost function (Case 2: when abatement is not possible)

- Secondary assumption (useful for when emissions depend on output directly)
 - Think: CO_2 emissions from coal-fired power plants.
- $C(x)$ is twice continuously differentiable, with $C' > 0$ and $C'' \geq 0$, and emissions depend on x according to $e = \delta(x)$, where $\delta' > 0$
- In some cases, we'll assume: $\delta(x) = \delta x$ (for $\delta > 0$)
 - This allows us to express profit as a function of emissions: $\Pi = p(e/\delta) - C(e/\delta)$

$$\implies \frac{d\Pi}{de} = \frac{p - C'(e/\delta)}{\delta}$$

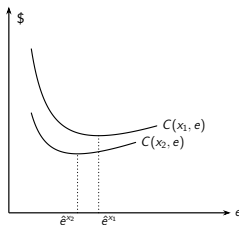
Competitive output markets

Firms' cost functions

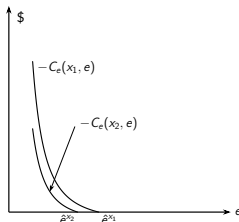
- Two output levels x_1 and x_2 , with $x_1 > x_2$.
- Case 1 (when abatement is possible)
 - Total costs are larger for greater levels of output.
 - When output increases, MAC shifts out.
- Case 2 (when abatement is not possible)
 - Unregulated optimal choice of e is associated with prices p_1 and p_2 for different levels of output.
 - When emissions are reduced by one unit:
 - Output $\downarrow$ by $1/\delta$
 - Costs $\downarrow$ by $c'(e/\delta)/\delta$
 - Revenue $\downarrow$ p_i/δ
- Trade-off b/w reduced costs and reduced revenue.

Case 1

Total costs

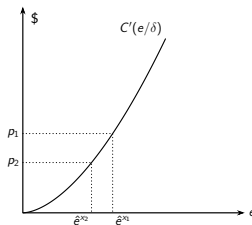


Marginal abatement costs

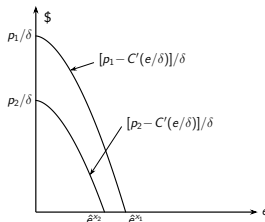


Case 2

Marginal costs



Marginal abatement costs



Market demand

- Households ($i = 1, \dots, I$) have quasi-linear utility:

$$U^i = u_i(x_i) + z_i - D_i(E)$$

- $U_i(x_i)$ is utility from consuming dirty good x
 - z_i is spending on all other goods
 - $D_i(E)$ is damage function
- Budget constraint: $y_i = px_i + z_i$
- Maximize utility s.t. budget constraint $\implies$ FOC: $u'_i(x_i) = p$
- Define i 's inverse demand (MWTP) for x as: $p_i(x_i) = u'_i(x_i)$
- Total benefit (i.e., consumer surplus) of consuming x is given by: $\int_0^{x_i} p_i(t) dt = u_i(x_i)$
- Aggregate $X = \sum_{i=1}^I x_i$, and let $P(X)$ define the inverse market demand for X

Welfare and social optimum

Case 1 (abatement is possible)

- Net benefits = difference between consumption utility (i.e., area under the inverse demand curve) and the private and environmental cost of its production.

$$W(x_1, \dots, x_J, e_1, \dots, e_J) \equiv \int_0^{\sum x_j} P(t) dt - \sum_{j=1}^J C^j(x_j, e_j) - D(E)$$

- Max $W(\cdot)$ w.r.t x_j, e_j :

$$\implies P(X) = C_{x_j}^j(x_j, e_j)$$

$$\implies D'(E) = -C_{e_j}^j(x_j, e_j), \text{ for } j = 1, \dots, J$$

Welfare and social optimum

Case 2 (abatement is not possible)

- Same process with altered cost function:

$$W(x_1, \dots, x_J) \equiv \int_0^{\sum x_j} P(t) dt - \sum_{j=1}^J C(x_j) - D(E) \text{ where } E = \sum_{j=1}^J \delta_j(x_j)$$

- Max $W(\cdot)$ w.r.t x_j (note: e_j are not choice vars):

$$\implies P(X) = C'_j(x_j) + D'(E)\delta'_j(x_j), \text{ for } j = 1, \dots, J$$

- With linear emissions ($e = \delta x$):

$$\implies P(X) = C'_j(x_j) + \delta D'(\delta X), \text{ for } j = 1, \dots, J$$

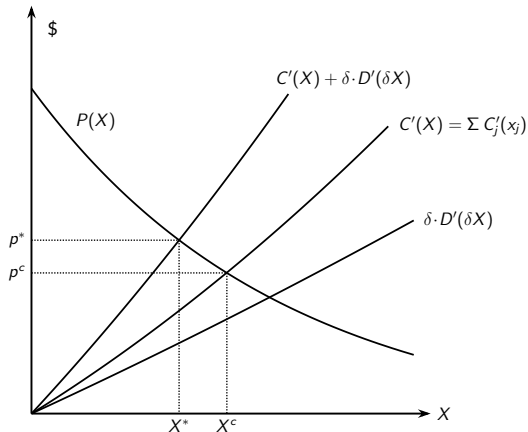
Welfare and social optimum

Case 2 (abatement is not possible)

- Linear emissions ($e = \delta x$):

$$\implies P(X) = C'_j(x_j) + \delta D'(\delta X) \quad \forall j$$

- This figure should look familiar!
- Social optimum:
 - $MB = MSC(\equiv MC + MED)$



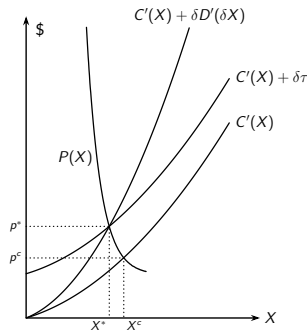
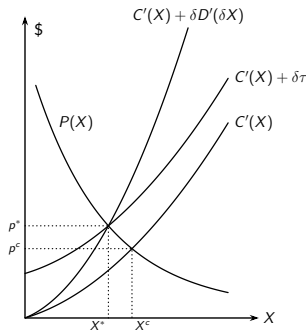
Do previous results hold (in the short run)?

- In the short run, yes! All previous results hold.
- Namely, that:
 1. There is an equivalence between emissions taxes and auctioned permits.
 2. Command-and-control will fail to achieve the social optimum (except in very isolated circumstances)
 3. Tradeable permits and subsidies can support the social optimum.

Emissions taxes

With competitive output markets...

- Two industries, identical cost structures.
- Different inverse demand curves (L: flatter, R: steeper)
- Tax is set at $\tau = D'(X^*)$
- Optimal allocation defined by intersection of demand and marginal social cost curve.
- Steepness of marginal cost and demand curves determines incidence of optimal tax.



- How does output, emissions, and prices change when we introduce/increase a tax?
 - i.e., $\partial X / \partial \tau < 0$, $\partial E / \partial \tau < 0$, $\partial p / \partial \tau > 0$.

Proposition 5.1

The introduction (or increase) of an emissions tax leads to:

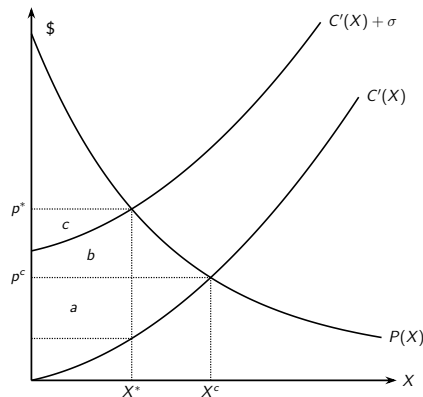
- (a) a decrease in both individual firms' (e_j) and aggregate emissions levels (E),
- (b) a decrease in individual firms' (x_j) and aggregate output levels (X), and
- (c) an increase in the output price (p_j).

(See Phaneuf and Requate (Ch. 5.2.1) for derivations).

Permits

Auctioned or freely distributed/transferable

- Regulator sets cap on emissions by distributing $L = E^*$ permits.
- Auctioning/trading results in eq'm permit price $\sigma(L)$.
- Firms' FOC:
 - $\Rightarrow p = C_{x_j}^j(x_j, e_j)$
 - $\Rightarrow \sigma = C_{e_j}^j(x_j, e_j)$, for $j = 1, \dots, J$
- If regulator chooses $L = E^*$, then:
 - $\sigma(L) = \sigma(E^*) = D'(E^*)$
 - And the output price is $p = P(X^*)$
- Auctioned vs. freely distributed $\rightarrow$ identical price change.
- If auctioned, scarcity rents (a+b+c) go to government.
- If distributed freely, rents remain with the industry.



Standards

Revisiting command-and-control

- Previously, we modeled an absolute level of emissions from firms, which is unrealistic
- Environmental policies often focus on relative standards (e.g., based on concentrations of pollutant)
 - Tailpipe auto emissions
 - NAAQS standards
 - Drinking water: e.g., Lead and Copper Rule (15ppb for lead or 1.3 ppm for copper)

Pollutant [links to historical tables of NAAQS reviews]	Primary/ Secondary	Averaging Time	Level	Form
Carbon Monoxide (CO)	primary	8 hours	9 ppm	Not to be exceeded more than once per year
		1 hour	35 ppm	
Lead (Pb)	primary and secondary	Rolling 3 month average	0.15 µg/m ³ ⁽¹⁾	maximum arithmetic mean of 3 consecutive monthly means in a 3-year period
Nitrogen Dioxide (NO₂)	primary	1 hour	100 ppb	Annual 98th percentile of 1-hour daily maximum concentrations, averaged over 3 years
	primary and secondary	1 year	53 ppb ⁽²⁾	Annual Mean
Ozone (O₃)	primary and secondary	8 hours	0.070 ppm ⁽³⁾	Annual fourth-highest daily maximum 8-hour concentration, averaged over 3 years
Particle Pollution (PM)	PM _{2.5}	primary	9.0 µg/m ³	annual mean, averaged over 3 years
		secondary	15.0 µg/m ³	annual mean, averaged over 3 years
	PM ₁₀	primary and secondary	35 µg/m ³	98th percentile, averaged over 3 years
		primary and secondary	150 µg/m ³	Not to be exceeded more than once per year on average over 3 years
Sulfur Dioxide (SO₂)	primary	1 hour	75 ppb ⁽⁴⁾	Annual 99th percentile of 1-hour daily maximum concentrations, averaged over 3 years
	secondary	1 year	10 ppb	annual mean, averaged over 3 years

Revisiting command-and-control

- Firms must comply with emissions standard: $e/x \leq \alpha$ (or $e \leq \alpha x$). Regulators choose α .
- ★ **Case 1:** if $e \leq \alpha x$ binds (i.e., $e = \alpha x$), we can substitute this into firms' cost function for e :

$$\Pi(x) = px - C(x, \alpha x)$$

$$\text{FOC: } \implies p = C_x(x, \alpha x) + \alpha C_e(x, \alpha x)$$

- The sol'n to this is given by a firms' output supply function: $x(\alpha)$ (i.e., how output changes with α)
- Regulator chooses emissions goal $\bar{e}$ and knows $x(\alpha)$, so chooses α to satisfy $\alpha = \bar{e}/x(\alpha)$.
- ★ **Case 2:** if $\alpha \geq \delta(x)$, then firm is in compliance. If not, firm must reduce x , perhaps to zero.

Absolute vs. Relative Standards

- If the regulator instead sets $\bar{e}$ directly (as we modeled previously), then firm maximizes $\Pi(x) = px - C(x, \bar{e})$.
- FOC: $\implies p = C_x(x, \bar{e})$, which defines the output supply response $x(\bar{e})$.
- If $\bar{e}$ is optimally chosen, the optimal solution will prevail.

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Comparing to Relative Standards

- | | |
|--|---|
| ○ $p - \alpha C_e(x(\alpha), \bar{e}) = C_x(x(\alpha), \bar{e}) > p$ | Rearranging relative standards FOC (where final $>$ follows from $-C_e > 0$) |
| ○ $p = C_x(x(\bar{e}), \bar{e})$ | But for absolute standards |
| ○ $C_x(x(\alpha), \bar{e}) > C_x(x(\bar{e}), \bar{e})$ | $\implies$ MC of x are larger for relative standards |
| ○ $\implies x(\alpha) > x(\bar{e})$ | Output response to regulation is strictly greater for relative standards |
| ○ $\implies C(x(\alpha), \bar{e}) > C(x(\bar{e}), \bar{e})$ | Total costs of meeting $\bar{e}$ are strictly greater for relative standards |
| ○ $\implies -C_e(x(\alpha), \bar{e}) > -C_e(x(\bar{e}), \bar{e})$ | MC of e are larger for relative standards |

Absolute vs. Relative Standards

Proposition 5.2

Suppose a regulator sets an emissions goal of $\bar{e}$ and uses a **relative standard** α to obtain it. If there is a single firm or J identical firms, the policy will lead to higher output, higher total costs, and higher marginal abatement costs relative to a comparable **absolute standard**.

Intuition:

- Firms can reduce emissions or *dilute* emissions by increasing output, so relative standards are less effective and more costly.
- Relative standards are only preferred when there are political/technical obstacles to implementing absolute standards or other feasible instrument.

Environmental policy in the long run

- Primary difference is that firms can enter/exit in the long run.
 - Assume symmetric firms (required by long-run zero profit condition)
- J identical firms with cost function that includes fixed (F) and variable (VC) costs:

$$C(x, e) = \begin{cases} VC(x, e) + F & x, e \neq 0 \\ 0 & x, e = 0 \end{cases}$$

- In LR, social optimum depends on optimal choice of x , e , and J .

Long-run social optimum

$$W(x, e, J) = \max_{x, e, J} \int_0^{x \cdot J} P(t) dt - J \cdot C(x, e) - D(e \cdot J)$$

- Where total output is $X = x \cdot J$ and total emissions is $E = e \cdot J$.
- FOCs that define optimal levels of x^* , e^* , and J^* :
 - $\implies P(X^*) = C_x(x^*, e^*)$
 - $\implies D'(E^*) = -C_e(x^*, e^*)$
 - $\implies P(x^*)x^* = C(x^*, e^*) + D'(E^*)e^*$
- Which can be rearranged:

$$P(x^*) = \frac{C(x^*, e^*) + D'(E^*)e^*}{x^*}$$

$\implies$ Firms will enter market up to point where market price reaches minimum average *social* cost curve.

What decides the long-run result?

- Free entry drives price to minimum average cost. The optimum needs price at minimum average *social* cost.
- So each instrument faces one question: **does the firm's average cost carry $D'(E^*)$?**
- **Taxes, auctioned permits.** The firm pays $D'(E^*)$ on every unit emitted. Damages enter average cost, and entry stops at J^* .
- **Freely allocated permits.** The allocation is lump-sum, so marginal conditions are untouched and J^* still obtains. Incumbents keep the rents.
- **Subsidies.** Payment for emitting below $\hat{e}$ is a lump-sum inflow. Average cost falls and entry overshoots.
- **Standards.** The cap binds each firm at e^* rather than the industry at E^* , so aggregate emissions scale with J .

Env. Policy in the long run: Standards

- Regulator wants to reach optimal emissions, E^* .
 - Sets $e = e^* = E^*/J^*$, where J^* is the optimal number of firms in the long run.
- Restricting emissions from each firm leads to greater SR profit.
 - Firms can charge higher prices for x .
 - This causes more firms to enter the market.
 - SR profits are kept rather than “given” to the environment.
- Implications:
 1. Industry size will be greater than socially optimal
 2. Each firm will produce less output than is socially optimal
 3. Total output and total emissions will be greater than socially optimal.

Env. policy in the long run: Taxes

- Regulator optimally chooses $\tau = D'(E^*)$.
- Standard marginal conditions ($p = C_x$ and $-C_e = \tau$) from profit maximization.
- Firms enter until market profits are zero, implying:

$$px - C(x, e) - \tau e = 0$$

- This is the LR “price = minimum average social cost” condition.

$\implies$ taxes achieve socially optimal levels of e^* , x^* , and J^* in the long run.

Env. policy in the long run: Auctioned permits

- Regulator optimally chooses $L = E^*$.
- Standard marginal conditions ($p(xJ) = C_x$ and $-C_e = \sigma$) from profit maximization.
- Firms enter until market profits are zero, implying:

$$p(xJ)x - C(x, e) - \sigma e = 0$$

- Which is the LR “price = minimum average social cost” condition.
- Demand for permits drives $\sigma = -C_e(x^*, e^*) = D'(E^*)$

$\Rightarrow$ auctioned permits achieve socially optimal levels of e^* , x^* , and J^* in the long run.

Env. policy in the long run: freely allocated, tradeable permits

- Regulator optimally chooses $L = E^*$ and allocates $\bar{e}$ to each incumbent firm
 - This cap applies existing firms and new entrants
 - Emissions cap cannot depend on number of new entrants
- Profit for incumbent firms: $\Pi(x, e) = px - C(x, e) - \sigma(e - \bar{e})$
- Profit for new entrants: $\Pi(x, e) = px - C(x, e) - \sigma e$
 - For $\sigma > 0$, both equations above cannot be zero.
 - New entrants will purchase permits from incumbent firms until $\Pi(x, e) = px - C(x, e) - \sigma e = 0$.
 - $\Rightarrow$ incumbent firms will sustain positive LR profits $= \sigma \bar{e} > 0$
 - $\Rightarrow$ incumbent firms who were initially allocated permits capture all scarcity rents.

$\Rightarrow$ **freely allocated permits achieve socially optimal levels of e^* , x^* , and J^* in the long run.**

Env. policy in the long run: Subsidies

- Regulator sets a per-unit subsidy ζ for emissions below $\hat{e}$
- Firms maximize: $\Pi(x, e) = P(X)x - C(x, e) - \zeta(e - \hat{e})$
- Optimal ζ must equal $D'(E^*)$.
 - $\implies P(X)x - C(x, e) - D'(E^*)(e - \hat{e}) = 0$ (after imposing LR profit condition)
 - Or, $P(X)x - C(x, e) - D'(E^*)e = D'(E^*)\hat{e}$
 - But, LHS must equal zero by FOC for optimal firm size
- Subsidy leads to a decrease in firms' fixed costs, which increases entry ($J > J^*$), aggregate output ($X > X^*$), and aggregate emissions ($E > E^*$)

$\implies$ **subsidies cannot achieve socially optimal levels of e^* , x^* , and J^* in the long run.**

Key takeaways

Instrument	In short run:	In long run:	Rents go to:
Emissions tax	optimal	optimal	government
Auctioned permits	optimal	optimal	government
Freely allocated permits	optimal	optimal	incumbents
Abatement subsidy	optimal	J, X, E too high	firms
Emissions standard $\bar{e}$	optimal [†]	J, X, E too high	firms

[†] With identical firms. A uniform standard across heterogeneous firms misallocates abatement.

- A relative standard α reaches the same $\bar{e}$ at higher output, higher total cost, and higher marginal abatement cost (Prop. 5.2).

Next class...

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- Tietenberg (1990) is outdated, but a nice policy/application companion to today's theory.
- Goulder & Parry (2008) is a great, less-technical treatment of this material.